



B.Tech. Degree III Semester Supplementary Examination November 2020/April 2021

ME 15-1304 FLUID MECHANICS
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain surface tension and capillarity.
- (b) What is meant by intensity of pressure? How it varies with depth of fluids.
- (c) Differentiate between the Eulerian and Lagrangian method of representing fluid motion.
- (d) Explain the terms 'metacentre' and 'metacentric height'.
- (e) What are the limitations of Bernoulli's equation?
- (f) Write a brief note on Pitot-tube.
- (g) Define velocity potential function and stream function.
- (h) Explain doublet and define the strength of doublet.
- (i) What is boundary layer?
- (j) Define Energy thickness.

PART B

(4 × 10 = 40)

- II. (a) An oil of viscosity 5 poise is used for lubrication between a shaft and sleeve. The diameter of the shaft is 0.5 m and it rotates at 200 r.p.m. Calculate the power lost in the bearing for a sleeve length of 100 mm. The thickness of oil film is 1 mm. (6)
- (b) What is a manometer? How are they classified? (4)
- OR
- III. (a) Define the equation of continuity. Obtain an expression for continuity equation for a three-dimensional flow. (8)
- (b) Mention the ways in which a flow pattern can be described. (2)
- IV. (a) Find the discharge of water flowing through a pipe 30 cm diameter placed in an inclined position where a venturimeter is inserted, having a throat diameter of 15 cm. The difference of pressure between the main and throat is measured by a liquid of specific gravity 0.6 in an inverted U-tube which gives a reading of 30 cm. The loss of head between the main and throat is 0.2 times the kinetic head of the pipe. (7)
- (b) What are the various losses of energy in pipes? (3)
- OR
- V. Derive the Hagen-Poiseuille equation for a laminar flow through a circular pipe. State the assumptions made. (10)

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- VI. (a) In a two-dimensional incompressible flow, the fluid velocity components are given by $u = x - 4y$ and $v = -y - 4x$. Show that velocity potential exists and determine its form. Find also the stream function. (6)

- (b) Explain the concept of circulation and vorticity. (4)

OR

- VII. (a) A free stream flow at 5 m/s is super imposed on a source placed at the origin. If the stagnation point occurs at $(-0.32, 0)$, make calculations for the strength of source, maximum width of Rankine half body and the flow velocity at point $P\left(0.5, \frac{\pi}{2}\right)$ in the flow field. (8)

- (b) What is Magnus effect? (2)

- VIII. (a) Determine the displacement thickness and momentum thickness in terms of the nominal boundary layer thickness δ in respect of the following velocity profiles in the boundary layer on a flat plate. (6)

$$(i) \quad \frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2$$

$$(ii) \quad \frac{u}{U} = \left(\frac{y}{\delta}\right)^{\frac{1}{m}}$$

Where u is the velocity at a height y above the surface and U is the free stream velocity.

- (b) Explain the formation of boundary layer on a flat plate. (4)

OR

- IX. (a) A plate 450 mm $\times$ 150 mm has been placed longitudinally in a stream of crude oil which flows with velocity of 6 m/s. Take specific gravity of crude oil is 0.925 and kinematic viscosity as 0.9 stoke. Calculate: (6)

(i) The thickness of the boundary layer at the trailing edge of the pipe.

(ii) Shear stress at the trailing edge.

(iii) The friction drag on the plate.

- (b) Briefly explain the boundary layer separation and its control. (4)
